	GLOBAL ECONOMICS **Grade: 10TH** **Learning Evidence 1** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective:	**Assessment criteria**
Structure decision-making problems under uncertainty, calculate and organise payoffs, and apply the Maximax and Maximin criteria to compare alternatives and justify decisions.	C1. Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision. C2. Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison. C3. Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison. C4. Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

Criteria assessment

This task sheet assesses criteria C1 through C4. Each problem assesses the criteria indicated in its title. The number of points assigned to each task is shown at the end of the item. Partial credit may be awarded when the correct procedure is shown but arithmetic errors are made. Answers that do not show the required procedure will not receive credit. A minimum of 4 points is required to pass each criterion.

Question	C1	C2	C3	C4
1	1	2	–	–
2	1	3	–	–
3	2	5	–	–
4	1	–	2	–
5	1	–	3	–
6	2	–	5	–
7	–	–	–	2
8	–	–	–	2
9	–	–	–	2
Total	**8**	**10**	**10**	**6**

1. Festival Beverage Booth [C1-1 C2-2]

A community festival company must decide which beverage booth to operate before knowing how strong customer attendance will be. The company can choose either a basic booth or a premium booth, while festival attendance may turn out to be high or low. If the company chooses the basic booth, it expects to earn a profit of sixteen thousand US dollars when attendance is high and ten thousand US dollars when attendance is low. If it chooses the premium booth, it expects to earn a profit of twenty-nine thousand US dollars when attendance is high and five thousand US dollars when attendance is low.

(C1) Characterize the key elements of the decision-making situation. **[1]**

(C2) Construct the payoff table. **[2]**

2. Local Delivery Fleet [C1-1 C2-3]

A local delivery firm must choose between a bicycle fleet and an electric-van fleet before traffic conditions are known. Traffic may be clear or congested. The bicycle fleet earns 4 thousand CNY per completed delivery batch and is expected to complete 14 delivery batches in clear traffic and 10 in congested traffic. The electric-van fleet earns 6 thousand CNY per completed delivery batch and is expected to complete 12 delivery batches in clear traffic and 8 in congested traffic. Clear traffic also adds a fixed scheduling bonus of 5 thousand CNY to either fleet, while congestion creates a fixed loss of 4 thousand CNY.

(C1) Characterize the key elements of the decision-making situation. **[1]**

(C2) Construct the payoff table. **[3]**

3. Regional Food Delivery Service [C1-2 C2-5]

A regional food delivery company must choose between a standard delivery service and an express delivery service before customer demand is known. Demand may be high or low. The standard service earns 3 thousand GBP per completed delivery order and also provides a fixed operating contribution of 5 thousand GBP. The express service earns 5 thousand GBP per completed delivery order and provides a fixed operating contribution of 8 thousand GBP. High demand adds another 2 thousand GBP per completed order to either service, while low demand adds 1 thousand GBP per completed order. The standard service is expected to complete 15 orders under high demand and 9 orders under low demand; the express service is expected to complete 11 orders under high demand and 7 orders under low demand.

(C1) Characterize the key elements of the decision-making situation. **[2]**

(C2) Construct the payoff table. **[5]**

4. Local Café Investment [C1-1 C3-2]

A small caf'e must decide whether to launch an artisan-dessert line or keep the current menu before foot traffic is known. Traffic may be high or low. All values are in million COP. Launching the dessert line produces revenue and cost of 58 and 16 under high traffic, and 27 and 19 under low traffic. Keeping the current menu produces revenue and cost of 49 and 21 under high traffic, and 41 and 21 under low traffic.

(C1) Characterize the key elements of the decision-making situation. **[1]**

(C3) Use revenue and costs to construct the payoff table. **[2]**

5. Custom Furniture Product Line [C1-1 C3-3]

A custom-furniture workshop must choose between producing dining tables and producing office desks before market conditions are known. Conditions may be strong demand with favourable material prices or weak demand with higher material prices. The respective quantities are 7 and 4 production lots for dining tables, and 6 and 3 lots for office desks. Revenue per lot depends on the product: 14 thousand EUR for dining tables and 16 thousand EUR for office desks. Cost per lot depends on the state: 6 thousand EUR under strong demand and favourable prices, and 9 thousand EUR under weak demand and higher prices.

(C1) Characterize the key elements of the decision-making situation. **[1]**

(C3) Use revenue and costs to construct the payoff table. **[3]**

6. Farm Subscription Plans [C1-2 C3-5]

A farm cooperative must choose between a standard produce box and a premium produce box before seasonal demand is known. Demand may be strong or weak. Quantities, in hundreds of subscriptions, are 8 and 5 for the standard box and 6 and 4 for the premium box under strong and weak demand, respectively. The plan-based revenue rates are 9 and 12 thousand USD per hundred subscriptions for the standard and premium boxes. Strong demand adds a state-based revenue rate of 3 thousand USD per hundred subscriptions; weak demand adds 1. The plan-based cost rates are 4 and 7 thousand USD per hundred subscriptions, respectively. The fixed distribution costs are 10 thousand USD under strong demand and 14 thousand USD under weak demand.

(C1) Characterize the key elements of the decision-making situation. **[2]**

(C3) Use revenue and costs to construct the payoff table. **[5]**

7. New Device Launch [C4-2]

A technology company must choose a strategy for launching a new wearable device. Market demand could be high, medium, or low. All payoffs are in thousands of dollars.

	Demand		
Alternative	High	Medium	Low
Premium launch	90	38	-25
Standard launch	65	45	20
Economy launch	42	34	27
Delay launch	18	18	18

(C4) What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

8. Restaurant Expansion [C4-2]

A restaurant chain must choose a strategy for expanding its operations. Customer demand could be high, medium, or low. All payoffs are in thousands of dollars.

	Customer demand		
Alternative	High	Medium	Low
Large expansion	85	32	-18
Moderate expansion	62	44	16
Small expansion	45	36	24
No expansion	20	20	20

(C4) What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

9. Renewable Energy Investment [C4-2]

A renewable-energy company must choose a strategy for developing a new solar project. Market conditions could be very favourable, favourable, stable, or unfavourable. All payoffs are in thousands of dollars.

S1 Very favourable market conditions.
S2 Favourable market conditions.
S3 Stable market conditions.
S4 Unfavourable market conditions.

	S1	S2	S3	S4
Large solar farm	110	55	10	-35
Medium solar farm	82	60	28	5
Small solar farm	58	48	35	18
Partnership project	45	40	32	25

(C4) What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**